

Time-of-Flight Mass Spectrometer

Teacher and presenter guide · Guided linear time-of-flight investigation · Murray State University

Purpose and learning target

Students analyze charged particles by timing their flight. They use three linked experiments to separate the effects of mass and positive charge, then discover why different ions with equal mass-to-charge ratio can produce one coincident detector peak.

Target claim: A time-of-flight detector measures arrival time directly. Under this ideal model, that time depends on mass-to-charge ratio, so one peak does not establish one kind of ion or identify a real molecule.

An **ion** is an electrically charged atom or molecule. The simplified path is: a voltage boost happens before timing, ions coast through a field-free tube in vacuum, and the detector records arrivals.

Units students see

Label	Meaning in this activity
dalton (Da)	A mass unit used for atoms and molecules.
elementary charge (e)	+1e or +2e is positive charge. The dimensionless charge number z is 1 or 2.
microsecond (μs)	One millionth of a second. The detector clock and horizontal trace axis show real flight time in μs .
metre (m)	Flight-tube length. Core investigations hold it at 1.50 m.
kilovolt (kV)	One thousand volts. Core investigations hold the boost at 20 kV.

The screen uses one scale for every core run: 1.00 physical μs takes 160 ms on screen, or 160,000 $\times$ slow motion. The detector trace keeps the same 0–18 μs horizontal axis.

12–15 minute core flow

- Orient (2 minutes).** Open `index.html`. Use the opening Guide overlay and identify the known simulated inputs, clock start, drift tube, and detector. Ask: “What will we know before Run, and what will the detector measure?”
- Investigation 1 — mass (3 minutes).** Both ions are +1. Students predict, then compare 100 Da and 400 Da on the shared axis.
- Investigation 2 — charge (3 minutes).** Both ions are 200 Da. Students predict, then compare +1 and +2.
- Investigation 3 — equal ratio (3 minutes).** Compare 200 Da/+1 with 400 Da/+2. Let the coincident arrival motivate the conventional m/z label.
- Transfer (2–4 minutes).** Ask whether one real detector peak proves one ion is present. Discuss what additional evidence a molecular identity would require.

Expected simulated evidence

Investigation	Center arrivals	Evidence-based explanation
Same charge; 100 vs 400 Da	Ion A: 7.64 μ s Ion B: 15.27 μ s	With charge and voltage held constant, the lighter ion crosses the tube sooner.
Same mass; +1 vs +2	Ion A: 10.80 μ s Ion B: 7.64 μ s	The +2 ion receives more kinetic energy from the same voltage boost and arrives sooner.
200 Da/+1 vs 400 Da/+2	Both: 10.80 μ s	$200 \div 1 = 200$ and $400 \div 2 = 200$. Both have $m/z = 200$, so the ideal detector records one coincident peak.

Keep inputs and evidence distinct. Sample cards show values the simulation already knows. Exact arrivals and the completed trace stay hidden until Run. The navy line is the summed detector signal; colored ticks identify the known packet centers. A peak represents a group arriving over a timing range.

Optional advanced extension: peak width

Introduce **timing width** before naming sigma. The extension compares 200 and 205 Da ions, both +1, in 1.00 m and 2.00 m tubes. It uses a shared 0–16 μs axis and an explicit classroom criterion: centers count as resolved when their separation is at least their mean full width at half maximum (FWHM).

Assumed width rule	1.00 m result	2.00 m result	What changes
Fixed $\sigma = 0.06 \mu\text{s}$	Separation 0.089 μs FWHM 0.141 μs overlapping	Separation 0.179 μs FWHM 0.141 μs resolved	Separation doubles while absolute width stays fixed.
$\sigma = 1.5\%$ of arrival time	Separation 0.089 μs mean FWHM 0.256 μs overlapping	Separation 0.179 μs mean FWHM 0.512 μs overlapping	Separation and width both double; their ratio stays constant.

The widths are assumptions, and this comparison does not support a universal promise that a longer tube improves resolution.

Model and boundaries

$$qV = \frac{1}{2}mv^2 \rightarrow t = L \sqrt{m / (2qV)}$$

- t is field-free flight time in seconds; L is length in metres; m is mass converted from Da to kilograms.
- $q = z \cdot e$ is positive charge in coulombs; z is the charge number; V is voltage in volts.
- The clock begins at drift-tube entry. Acceleration time is excluded.
- The Gaussian peak uses σ as its timing spread, $\text{FWHM} = 2.355\sigma$, and area proportional to packet count.

This ideal linear TOF does not model an ion source, extraction optics, reflectron focusing, detector efficiency, calibration, fragmentation, uncertainty, or molecular identification.

Discussion prompts

- Which values are controlled inputs, and which values are detector observations?
- Why does the +2 ion arrive first when mass is held constant?
- How can two different ions make one peak in the equal-ratio experiment?
- What additional evidence would you need before identifying a real molecule?
- In the extension, which assumed width rule lets the longer tube cross the stated criterion, and why?

Controls: ? reopens the Guide. ⚙ Settings offers Open Presenter Notes (this document), Presentation mode, and Reset. Run locks the prediction during flight; Replay uses the same physical-to-screen scale.

Sources and attribution

Iowa State University Chemical Instrumentation Facility, [Mass Spectrometry Tutorial](#), for qV acceleration, field-free drift, and detector timing. Shimadzu, [MALDI basics](#), for pulsed-ionization context. A Murray State project record supports plausible course fit; it is not named here. These sources do not endorse this activity, and the demo is not presented as any instructor's instrument or assignment.

